

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	16 February 2026
Team ID	LTVIP2026TMIDS63806
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Integration & Preparation	• Connect Tableau/Power BI to the heart disease dataset(s). • Clean and format patient data (handle missing values, correct data types, standardize age groups, etc.). • Create calculated fields (e.g., Risk Percentage, Age Groups, Cholesterol Categories, Blood Pressure Levels).
FR-2	Core Visualizations Development	• Build demographic trend charts (e.g., age & gender distribution of heart disease cases). • Develop risk factor comparison charts (e.g., cholesterol vs blood pressure correlation). • Create category-wise breakdowns (e.g., chest pain type, fasting blood sugar, smoking habits).
FR-3	Dashboard Interactivity & User Controls	• Implement global filters (e.g., age group, gender, cholesterol range). • Add dropdown filters for risk categories and clinical indicators. • Enable interactive tooltips to show detailed patient metrics on hover.
FR-4	Storytelling & Deployment	• Combine visualizations into a cohesive Heart Disease Analysis Dashboard or Story. • Publish the dashboard to Tableau Public/Server or Power BI Service. • Enable export options (PDF/Image) for reporting and presentations.
FR-5	Comparative & Risk Analysis	• Provide comparison between high-risk and low-risk patient groups. • Generate visual highlights for high-risk demographics (e.g., specific age groups or gender).
FR-6	Trend & Pattern Detection	• Identify and display key health trends (e.g., increasing cholesterol levels across age groups). • Implement visual indicators (color coding, reference lines) to highlight abnormal or critical values.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard must be intuitive and easy to navigate. It should use clear color coding for risk levels (e.g., high, medium, low), readable fonts, and self-explanatory filters so both medical and non-technical users can explore patient data easily.
NFR-2	Security	If deployed internally, role-based access control or Row-Level Security (RLS) must restrict access to sensitive patient data. If published publicly for demonstration, patient-identifiable information must be removed or anonymized.
NFR-3	Reliability	Visualizations must accurately reflect the underlying patient dataset. Data refresh processes must not distort or corrupt health metrics.
NFR-4	Performance	The dashboard should load quickly (ideally within 3–5 seconds). Filters and interactions must update visualizations smoothly without noticeable delay.
NFR-5	Availability	The dashboard should be accessible 24/7 to authorized users via Tableau Server, Tableau Cloud, or Power BI Service.
NFR-6	Scalability	The data model should support future expansion (e.g., adding more patient records, additional risk factors, or longitudinal time-series data) without requiring a complete redesign or affecting performance.